

3.4 Management of threats to tropical rainforests

Tropical rainforests offer a number of resources for human use. These include:

- **Medicinal plants:** many medicines derive from plants found in the tropical rainforest. For example, quinine to treat malaria and cortisone to treat inflammation and allergies.
- **Cash crops:** farmers grow crops such as coffee, maize, and cocoa for money.
- **Minerals and mining resources:** coal, oil, gold, silver, and many more minerals are mined in the tropical rainforests.
- **Wood products:** hardwood trees are harvested for their different wood types. The wood is used for many different purposes, including flooring and furniture.
- **Ecological services:** tropical rainforests offer benefits which positively support the wellbeing of humanity. For example, food, clean water, clean air, flood and erosion control, biodiversity, and educational and recreational opportunities.

However, as with any resources, the uncontrolled harvesting of resources from rainforests poses a threat to them and the economies that rely on them (Table 3.4). This results in **deforestation** and **rainforest degradation**.

Driven by demand for palm oil in food and industrial demand, global oil palm production has more than doubled over the last 20 years. Between 1997 and 2018, palm oil plantations have expanded from 100 thousand km² to 210 thousand km², and the production of palm oil has increased from 100 million tons to 300 million tons. This has resulted in the deforestation of large areas of tropical rainforest.

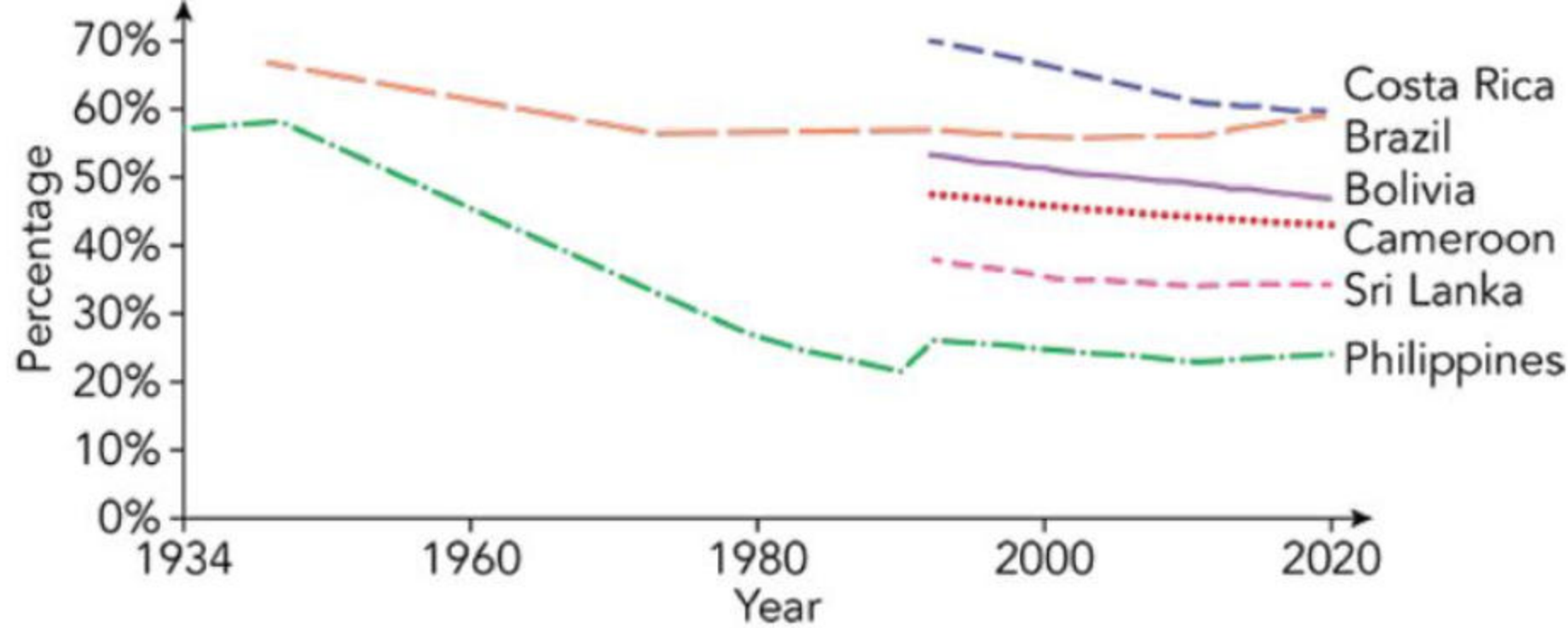
Source A: Palm oil plantation expansion

TASK 1: Study Source A

- Calculate the percentage increase** in the area under palm oil plantations between 1997 and 2018.
- Explain why planting oil palms in place of tropical rainforest decreases biodiversity.
- In pairs, discuss the sustainable use of rainforests. What do you think could be done to protect the tropical rainforests that are being felled for palm oil, while still protecting the palm oil industry? Write a list of expanded bullet points, giving your main ideas.

Table 3.4: The threats to tropical rainforests

Factor	Threat
Medicinal plant extraction	Overharvesting of specific plant species for medicinal purposes results in their numbers decreasing. This threatens the species and can impact the wider food chain.
Agriculture expansion	The expansion of cattle and crop farming requires the clearing of forested areas. Farming is responsible for up to 90% of the global loss of tropical rainforests. Farmers often use chemicals, such as fertilisers and pesticides, that are harmful to ecosystems.
Urbanisation	The growing human population means an increased need for housing, which leads to the clearing of rainforests for human settlements. For example, to make space for the construction of buildings and roads.
Timber harvesting or logging	Some of the most valuable hardwood timbers, such as teak and rosewood, come from rainforests. Areas of rainforest are felled to obtain the wood from the trees.
Hydroelectric power production	Rivers in the tropical rainforests are dammed to create hydro-electric power. This changes the flow of the rivers and floods areas upstream.
Mining and mineral extraction	Mining for minerals requires the felling of rainforests to clear the area for excavation activities and for transport routes.
Climate change	Global emissions of greenhouse gases, such as carbon dioxide (CO ₂), cause climate change by increasing temperatures and decreasing rainfall. This is a threat to the rainforests that depend on constant temperatures and rainfall for healthy vegetation growth.



Source B: Global rates of deforestation shown as a decrease in percentage of land covered by tropical rainforest

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TASK 2: Study Source B

- Identify** the country with an increase in the share of land covered by rainforest from 2010.
- Why do you think some countries only show information starting in 1990?
- Describe the change** in the share of land covered by rainforest for the Philippines.
- Describe the trend** for rainforest cover for most countries in the graph.

The uncontrolled use of resources from tropical rainforests is unsustainable. It causes long-term damage to this vital ecosystem. Deforestation, the intentional clearing of forested areas, is the most significant impact of resource extraction. The destruction of forested land has a number of impacts on the ecosystems and the people and economies relying on it. These include:

Soil erosion

As vegetation is removed, it can no longer protect the soil from rain and running water. Water washes the soil away, which prevents plants from growing and habitats from developing, and negatively impacts biodiversity.

Loss of biodiversity

The removal of vegetation results in the loss of habitat for all rainforest organisms. This results in a decline in the number of species in the area. Loss of biodiversity can also be linked to the extraction of plants for medicinal purposes or trees with a specific type of wood. The targeting of specific species results in their numbers declining to the point of potential extinction.

Water scarcity

The felling of trees interrupts the water cycle. Fewer trees means that less water is taken up by tree roots and released into the atmosphere via transpiration. This leads to less cloud formation and rainfall decreases. This impacts water availability both locally and regionally.

Loss of resources for the Indigenous population

Indigenous populations have historically relied on the sustainable use of rainforest resources for survival, not only for medicines, but also for shelter and food. In modern times, the unsustainable demand for resources from outside parties has increased. This has resulted in conflict between those taking resources and the communities needing to source their basic needs from them.

Increased risk of global climate change

With the removal of trees, less carbon dioxide is absorbed from the atmosphere during photosynthesis and less carbon is stored in trees' structures. This adds to the challenge of global climate change.

Environmental pollution from oil mining

The unmanaged release of oil into the environment results in the pollution of surface and ground water and soils, which is harmful to plant and animal life. This affects the health of ecosystems.

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Reduced ecological services

Deforestation impacts the ecological services that rainforests offer. It reduces their ability to clean water and air, and supply food and other resources that

humans rely on.

Hydroelectric plant construction

This has several impacts on the river, including:

- stopping the natural flow of the rivers
- blocking off migratory routes of aquatic species
- changing river temperature and water clarity
- depriving downstream areas of essential flood water.

Economic issues

All products that are extracted from the rainforest financially support people on a local, regional, and even international level. For example, local jobs depend on the harvesting of products from the rainforest, while businesses such as construction, energy producers, and pharmaceutical companies rely on products harvested in the rainforest. If these products are unsustainably harvested, these individuals and companies will be negatively impacted as demand begins to outstrip supply.



Source **C**: An aerial view of an iron ore mine in Para State, Brazil

TASK 3: Study Source **C**

a

Draw an annotated sketch diagram of the iron ore mine in Para State, Brazil.

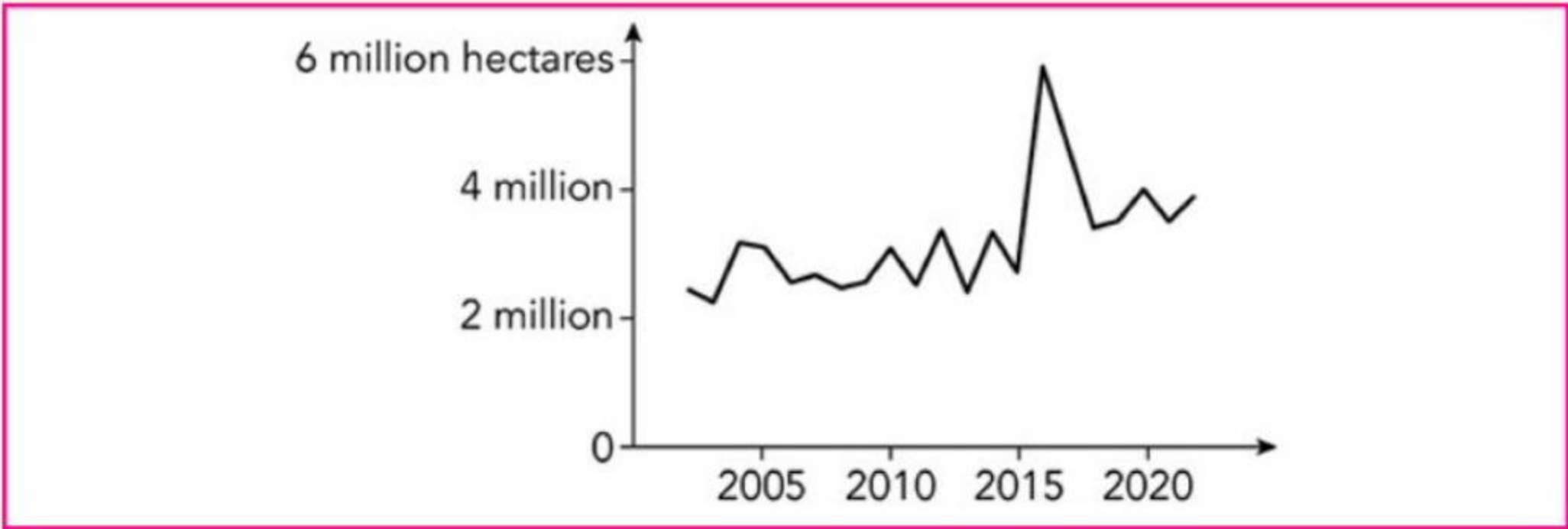
b

Describe how deforestation and heavy rainfall could affect a steep slope like this.

c

Healthy rainforest topsoils have nutrients that plants use for growth. With a partner, discuss how the removal of the topsoils will make rehabilitation of this site more difficult. Note down your answer and share the information with another pair.

IMPROVE THIS ANSWER



Source **D**: Rates of deforestation in the tropical rainforests between 2000 and 2023

Question 1: Describe how the rates of deforestation change between 2000 and 2023.

Answer 1:

The rate of deforestation increases from the start of the graph to the end. In 2016 there was a sudden increase in the rate of deforestation, then it decreases again.

Question 2: Explain some causes for the changes in rates of deforestation identified in question 1.

Answer 2:

Deforestation is taking place because humans use the resources from the rainforests. The trees and land are used for different reasons. Because there are more people there is an increase demand on the resources so deforestation is increasing.

Both of these answers require improvement. Rewrite the answers improving them with the guidance tips given below:

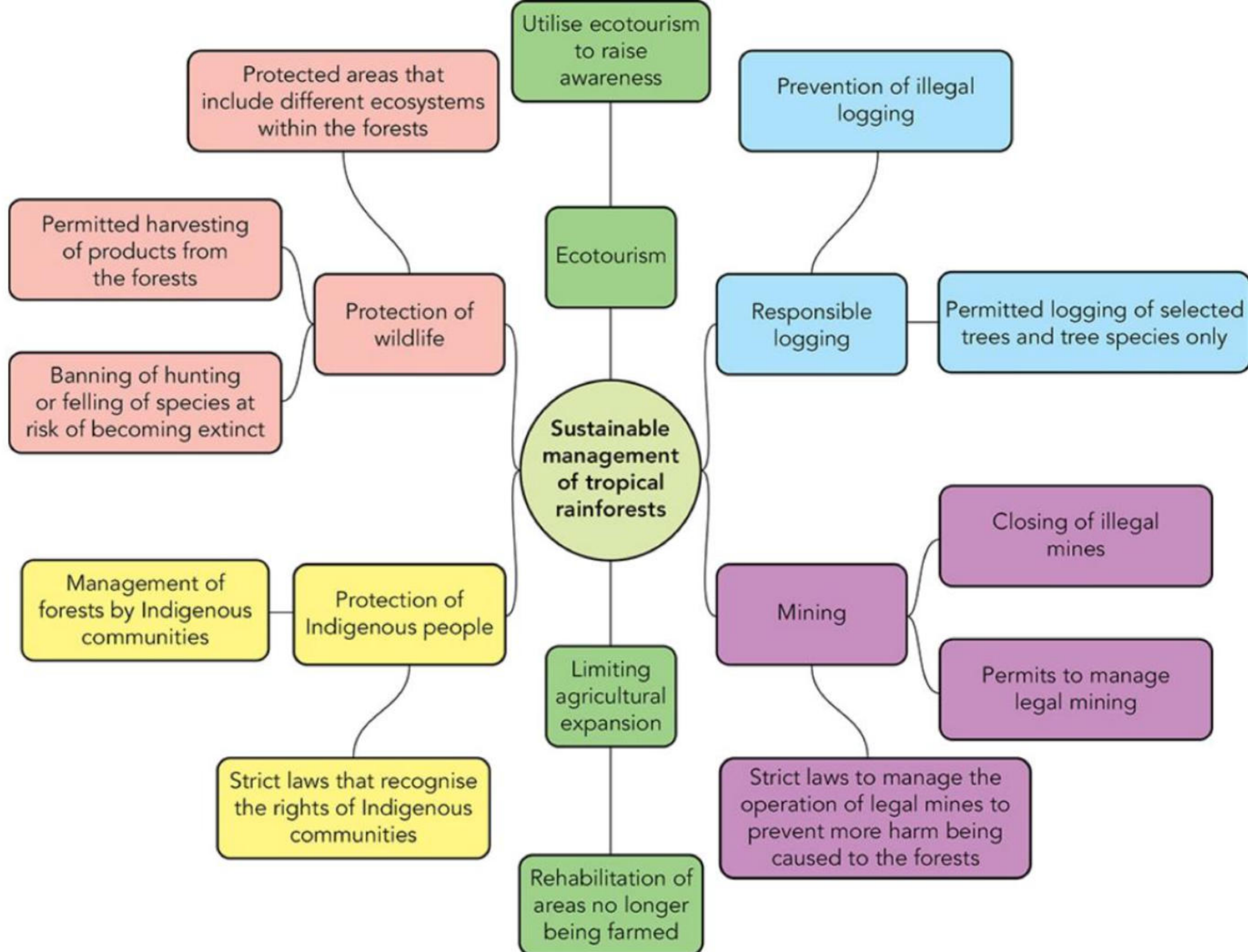
- Identify the command word.
- What type of answer is the command word seeking?
- Does the answer given correctly respond to the command word and answer the question?
- Has the answer included data where necessary?
- Has relevant terminology been used in the answer?

Sustainable management of tropical rainforest ecosystems

There are many strategies and techniques that humans can employ to prevent ongoing damage to tropical rainforest ecosystems (Source E). The strategies used include international and national policies that try to protect forests in the long-term, while techniques can include the use of ideas like selective logging, which identifies specific trees that can be felled while protecting the rest of the forest around them. However, growing human populations means that there is an ever-growing demand for products from the rainforest, and many people rely on the income from those products for survival. Globally, tropical rainforests are still under extreme pressure from human demand. Ongoing awareness and changes in human behaviour is essential for the true protection of these important ecosystems.

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The success of these strategies and techniques are limited by political and individual willingness to protect forested areas. These forests are seen as an economic opportunity; by not harvesting wood or resources from the tropical rainforest, those regions may suffer greater economic hardship. In addition, the decision on what happens with any forested area is up to the country in which the forest exists. Therefore, international laws to protect these regions are more difficult to achieve. Countries are working to try and find ways to support economic growth in these countries while protecting the forests, but large regions of forest are still lost each year.



Source **E**: Sustainable management of tropical rainforest ecosystems

TASK 4: Study Source **E**

- Write a 250 to 350-word summary of how ecotourism, responsible logging, and the protection of wildlife help to sustainably manage tropical rainforests.
- Think of one other method (other than those given in part a) that could be used to help manage the tropical rainforest ecosystem. Refer to the threats in Table 3.4 to guide you.
- In pairs, discuss the factors shown on the mind map. Consider the challenges faced by governments or communities when trying to put these ideas in place. Write down four challenges and share them with your class.

THINK LIKE A GEOGRAPHER

Geographers need to consider changes at different scales. Why is the burning of fossil fuels and the resulting release of pollutants and greenhouse gases affecting ecosystems locally, while also affecting tropical rainforests many thousands of km away?